A DISTINGUISHED TRIANGLE FROM A PUSHOUT SQUARE OF PRESHEAVES OF SETS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the reduction and the triangle are both inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Sheafification of presheaves of sets and amalgamated sums. Same chapter, the sheafification of a presheaf of sets and the description of an amalgamated sum of sheaves in terms of a pushout of presheaves. Supplies hypotheses (1) and (2) and the objects whose cohomology is compared.
- Cohomology of a presheaf of sets as discussed in the section on lim-presheaves. Same chapter, the cohomology RGamma(K, K) attached to a presheaf of sets, defined through the topos-theoretic construction of the section. Supplies the functors appearing in the distinguished triangle.

Source reference. Original section 079X: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2017–2017. Navigation pointer invoked at proof line 6149; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03CI: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6159–6159. Navigation pointer invoked at proof line 6159; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0EVY: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6074–6074. Navigation pointer invoked at proof line 6167; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Consider a commutative diagram*

$$\begin{array}{ccc} \mathcal{D} & \longrightarrow & \mathcal{F} \\ \downarrow & & \downarrow \\ \mathcal{E} & \longrightarrow & \mathcal{G} \end{array}$$

of presheaves of sets on $\mathcal{C}$ and assume that

(1) *$\mathcal{G}^\# = \mathcal{E}^\# \amalg_{\mathcal{D}^\#} \mathcal{F}^\#$, and*
(2) *$\mathcal{D}^\# \to \mathcal{F}^\#$ is injective.*

Then there is a canonical distinguished triangle

$$R\Gamma(\mathcal{G}, K) \to R\Gamma(\mathcal{E}, K) \oplus R\Gamma(\mathcal{F}, K) \to R\Gamma(\mathcal{D}, K) \to R\Gamma(\mathcal{G}, K)[1]$$

functorial in $K \in D(\mathcal{C})$ where $R\Gamma(\mathcal{G}, -)$ is the cohomology discussed in Section 079X.

Proof. Since sheafification is exact and since $R\Gamma(\mathcal{G}, -) = R\Gamma(\mathcal{G}^\#, -)$ we may assume $\mathcal{D}, \mathcal{E}, \mathcal{F}, \mathcal{G}$ are sheaves of sets. Moreover, the cohomology $R\Gamma(\mathcal{G}, -)$ only depends on the topos, not on the underlying site. Hence by Sites, Lemma 03CI we may replace $\mathcal{C}$ by a "larger" site with a subcanonical topology such that $\mathcal{G} = h_X$, $\mathcal{F} = h_Y$, $\mathcal{E} = h_Z$, and $\mathcal{D} = h_E$ for some objects X, Y, Z, E of $\mathcal{C}$. In this case the result follows from Lemma 0EVY. □